MATH317 · NUMERICAL METHODS

Chapter 3.2 Newton's Method

Midterm summary · Lecture 6

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01 What is Newton's method?

A method for finding a root of `f(x) = 0`. It is also called the **Newton-Raphson iteration**. It is one of the most important procedures in numerical analysis and is much faster than bisection.

Requirement

The function must be **Differentiable**, because the method needs `f'(x)`.

The formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Start from a guess x_0 and substitute to get x_1. Substitute x_1 to get x_2, and so on.

02 Where does the formula come from?

First view: the tangent line

At the point $(x_0, f(x_0))$ draw the tangent. Near that point the tangent is close to the curve, and its equation is:

$$l(x) = f'(x_0)(x - x_0) + f(x_0)$$

Instead of the root of the hard function, find the root of the tangent. Set $l(x) = 0$ and solve:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Second view: Taylor series

Ask: what correction h added to x_0 gives the root? We want $f(x_0 + h) = 0$.

$$f(x_0 + h) = f(x_0) + h\, f'(x_0) + \frac{h^2}{2!} f''(x_0) + \ldots = 0$$

Keep the first two terms only: $f(x_0) + h\, f'(x_0) = 0$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Both views give the same formula.

03 The slide example

If $f(x) = x^3 - x + 1$ and $x_0 = 1$, what are x_1 and x_2 in the Newton iteration?

1 Differentiate

$f(x) = x^3 - x + 1$ $\qquad$ $f'(x) = 3x^2 - 1$

2 Compute x_1

$f(1) = 1 - 1 + 1 = 1$ $\qquad$ $f'(1) = 3 - 1 = 2$

$$x_1 = 1 - \frac{1}{2} = \frac{1}{2}$$

3 Compute x_2

$$f\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{1}{2} + 1 = \frac{5}{8} \qquad f'\left(\frac{1}{2}\right) = \frac{3}{4} - 1 = -\frac{1}{4}$$

$$x_2 = \frac{1}{2} - \frac{\frac{5}{8}}{-\frac{1}{4}} = \frac{1}{2} + \frac{5}{2}$$

$$x_2 = 3$$

Notice

Here $x_2 = 3$ jumped far from x_0. Newton's method does not guarantee convergence when the first guess is far from the root.

04 An example showing Newton's speed

The slides solve $x^3 + x = 2x^2 + 3$. Move everything to one side:

$$f(x) = x^3 - 2x^2 + x - 3 = ((x - 2)x + 1)x - 3$$

$$f'(x) = 3x^2 - 4x + 1 = (3x - 4)x + 1$$

Both are written in **Nested form** so they are cheaper to evaluate. Start from $x_0 = 3$:

n	x_n	$f(x_n)$
0	3.0	9.0
1	2.4375	2.04
2	2.21303	0.256
3	2.17555	6.46×10^{-3}
4	2.17456	4.48×10^{-6}
5	2.17456	2.2×10^{-12}

The real root is `2.17456`. Watch how fast `f(xₙ)` shrinks: the number of correct digits roughly doubles every step.

05 Convergence analysis

Let the error at step n be `eₙ = xₙ − r`, where `r` is the root.

Theorem 1

The errors of Newton's method are related by:

$$e_{n+1} = -\tfrac{1}{2} \cdot \frac{f''(\xi_n)}{f'(x_n)} \cdot e_n^2 \qquad (\xi_n \text{ between } x_n \text{ and } r)$$

$$|e_{n+1}| \le c \cdot |e_n|^2$$

The new error is proportional to the square of the old one. This is called **Quadratic convergence**.

If the error is now	After one step it is about
10^{-2}	10^{-4}
10^{-4}	10^{-8}

Condition

This holds only when x_0 is sufficiently close to the root.

06 When do we stop?

After each step compute the **Absolute relative approximate error**:

$$|\varepsilon_a| = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| \times 100$$

1 **Compute $|\varepsilon_a|$.**

2 **Compare it with the given tolerance ε.**

3 **If $|\varepsilon_a| > \varepsilon$ continue. Otherwise stop.**

Also stop when the number of iterations exceeds the allowed maximum.

The pseudocode on the slide

It starts from a value `x` and runs at most `n_max` iterations. It needs two functions, `f(x)` and `f'(x)`. The parameters ε and δ control convergence and accuracy.

07 Newton vs Bisection

	Bisection	Newton
What it needs to start	An interval `[a, b]` with a sign change	One guess x_0
Needs the derivative?	No	Yes
Speed	Slow: the error halves each step	Fast: **Quadratic convergence**
Is convergence guaranteed?	Yes	No, unless x_0 is close to the root
Work per step	One evaluation: `f(c)`	Two evaluations: `f` and `f'`

08 How it shows up in the midterm

Newton's method appeared in one of the four midterms (461), for five marks.

> Given $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, apply Newton's method for two iterations.

$$f'(x) = 3x^2 - 2$$

$$f(1.5) = 3.375 - 3 - 1 = -0.625 \qquad f'(1.5) = 6.75 - 2 = 4.75$$

$$x_1 = 1.5 - \frac{-0.625}{4.75} = 1.5 + 0.1316$$

$$\mathbf{x_1 = 1.6316}$$

$$f(1.6316) = 0.0803 \qquad f'(1.6316) = 5.9864$$

$$x_2 = 1.6316 - \frac{0.0803}{5.9864} = 1.6316 - 0.0134$$

$$\mathbf{x_2 = 1.6182}$$

If the question also asks for the error between the last two values:

$$|\varepsilon_a| = \left|\frac{1.6182 - 1.6316}{1.6182}\right| \times 100 = 0.83\ \%$$

09 Common mistakes

Mistake	Fix
Adding instead of subtracting in the formula	$x_{n+1} = x_n - \frac{f}{f'}$. Watch the sign when f is negative.
Differentiating incorrectly	Write $f'(x)$ on the first line and check it.
Substituting x_0 again in the second step	The second step uses x_1.
Rounding x_1 too much before continuing	Keep at least four decimal places.
Saying Newton always converges	No. It needs x_0 close to the root.

10 Check yourself

1. $f(x) = x^2 - 2$ and $x_0 = 1$. Find x_1 and x_2.
2. $f(x) = x^3 - x - 1$ and $x_0 = 1.5$. Find x_1 and x_2.
3. What does quadratic convergence mean?

Solutions

1) $f'(x) = 2x$

$$x_1 = 1 - \frac{1 - 2}{2} = 1 + 0.5 = 1.5$$

$$x_2 = 1.5 - \frac{2.25 - 2}{3} = 1.5 - 0.0833$$

$\mathbf{x_2 = 1.4167}$ (true root $\sqrt{2} = 1.4142$)

2) $f'(x) = 3x^2 - 1$

$f(1.5) = 0.875$ $f'(1.5) = 5.75$

$$x_1 = 1.5 - \frac{0.875}{5.75} = 1.3478$$

$f(1.3478) = 0.1007$ $f'(1.3478) = 4.4499$

$$\mathbf{x_2 = 1.3478 - \frac{0.1007}{4.4499} = 1.3252}$$

3) The new error is proportional to the square of the old one: $|e_{n+1}| \le c\,|e_n|^2$. The number of correct digits roughly doubles each step.